A REMARK ON OPENAI FAMILY 176: THE CONSTANT IN THE SECOND KAHN–KALAI BOUND

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Abstract. OpenAI's manuscript *The second Kahn–Kalai conjecture* (family 176 of `github.com/openai/math`, commit `adc7f124`) proves $p_c(n,H) \le \min\{1, 2048e^{50} p_E(n,H)(1+\log_2 h)\}$ and $p_c(n,H) \le 6144e^{50} p_E(n,H)\log_2 n$. We run the same proof with its numerical choices left as parameters, and sharpen three elementary estimates. The result is $p_c(n,H) \le \min\{1, 10^5\, p_E(n,H)(1+\log_2 h)\}$ and $p_c(n,H) \le 2\cdot 10^5\, p_E(n,H)\log_2 n$. This improves the constants by factors of about $1.06\cdot 10^{20}$ and $1.59\cdot 10^{20}$. Nearly all of the gain, a factor $e^{43.85} \approx 1.1\cdot 10^{19}$, comes from the margin $a/\rho = e^{-50}$ in the local resampling lemma, which the proof does not need. The method is OpenAI's; this note only does the bookkeeping.

1. Statement

We use the notation of the manuscript: $p_c(n,H)$ is the least p with $\mathbb{P}(N(G(n,p),H)>0) \ge \frac12$, and $p_E(n,H)$ is the least p at which every $F \subseteq H$ has $\mathbb{E}N(G(n,p),F) \ge \frac12$ (`introduction.tex` L17–23; the Lean statement `SecondKahnKalai.lean` uses the same $\frac12$).

Theorem 1. *For every $n \ge 2$ and every graph H with $h \ge 1$ edges and at most n vertices,*

$$p_c(n,H) \le \min\{1,\ C_1\, p_E(n,H)(1+\log_2 h)\}, \qquad p_c(n,H) \le C_2\, p_E(n,H)\log_2 n,$$

with $C_1 = 10^5$ and $C_2 = 2\cdot 10^5$. The certified values are $C_1 \le 99\,379$ and $C_2 \le 197\,985$.

Fix the parameters

$$L = \tfrac{123}{20}, \quad \alpha = \tfrac75, \quad \beta = \tfrac25, \quad g = 6, \quad \lambda = \frac{(g+1)^{g+1}}{g^g} = \frac{7^7}{6^6}, \quad x = e^{-L}, \quad c = \alpha+\beta+\log(1+x).$$

In the manuscript, $L = 50$, $\alpha = 10$, $\beta = 1$, $g = 16$ and the spread multiplier is 128 instead of λ.

2. The proof with parameters

Each step below is the manuscript's argument with its numbers replaced by parameters. We point to the line that changes.

Lemma 2 (Hierarchy; `graphs.tex` L50–112)**.** *Put $q = p_E(n,H)$ and $\sigma = \lambda q$. There is a disjoint probability tree whose path unions are copies of H, with every level σ-spread and label sizes $1 \le \ell_1$, $\ell_{i+1} \ge g\ell_i$, $\ell_k \le h$.*

Proof. The argument is the manuscript's, with 17 replaced by $g+1 = 7$. It needs $2\binom{h_I}{r}\lambda^{-r} < 1$ whenever $r \ge h_I/7$ and $r \ge 1$. Since $\binom{h_I}{r} \le \binom{7r}{r}$, it suffices that $2\binom{7r}{r} < \lambda^r$ for all $r \ge 1$. Robbins' form of Stirling's formula [3] gives $\binom{n}{k} \le \sqrt{n/(2\pi k(n-k))}\, n^n/(k^k(n-k)^{n-k})$. With $n = 7r$, $k = r$ this is $\sqrt{7/(12\pi r)}\,\lambda^r$, and $2\sqrt{7/(12\pi)} < 0.87$. We also checked the inequality in exact integer arithmetic for $r \le 3000$. Maximality then gives $|S| < |I|/7$, so $\ell_{i+1} > 6|H_i| \ge 6\ell_i$, and the conditional-spread computation (L94–112) is unchanged. □

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Lemma 3 (Local resampling; `resampling.tex` L59–148)**.** *Suppose $|A_b| \le m$, ν is a-spread, $0 < \rho < 1$ and $a/\rho \le x$. Call a transition a failure if $Z(Y) < \mathrm{e}^{-\alpha m}$ or $|T| > m/2$. Then*

$$\mathbb{P}(\text{failure}) \le \delta(m) := \mathrm{e}^{-\alpha m} + \mathrm{e}^{\alpha m} \sum_{r > m/2} \binom{m}{r} x^r, \qquad \mathbb{E}[\mathbf{1}_{\{\text{target}=b,\, Z \ge \mathrm{e}^{-\alpha m}\}} f(W)] \le \mathrm{e}^{\alpha m} (1+x)^m \nu_b \, \mathbb{E} f(W).$$

Proof. This is the manuscript's proof with 10 replaced by α. We keep the two exact sums and drop the final rounding to e^{-9m} and e^{11m}. □

Lemma 4 (Reduction; `trees.tex` L56–205)**.** *Suppose the capacities satisfy $m_{j+1} \ge g\, m_j$, every level is a-spread, and $\rho = \mathrm{e}^L a < 1$. Use pruning threshold $\epsilon_j = \mathrm{e}^{-\beta m_j}$. Then one attempt succeeds with probability at least $1 - B$, where*

$$B = \sup_{m \ge 1} \varphi_0(m) + \sum_{j \ge 2} \sup_{m \ge g^{j-1}} \varphi_0(m)\, \mathrm{e}^{cS(m)}, \qquad \varphi_0(m) = \mathrm{e}^{\beta m} \delta(m), \quad S(m) = \sum_{i \ge 1} \lfloor m/g^i \rfloor.$$

On success, level j becomes $\alpha_j/(1 - \mathrm{e}^{-\beta m_j})$-spread and its capacity becomes $\lfloor m_j/2 \rfloor$.

Proof. The recurrence (L109) becomes $D_v = \mathrm{e}^{\beta m_s}(d_v + \sum_b t_{vb} D_b)$. Lemma 3 then gives $\mathbb{E}D_v \le \varphi_0(m_s) + \mathrm{e}^{cm_s} \sum_b \nu_v(b) \mathbb{E} D_b$, so $\mathbb{E}D_o \le \sum_j \mathrm{e}^{c\sum_{h<j} m_h} \varphi_0(m_j)$. Since m_{j-i} is an integer at most m_j/g^i, we have $m_{j-i} \le \lfloor m_j/g^i \rfloor$, hence $\sum_{h<j} m_h \le S(m_j)$. Also $m_j \ge g^{j-1}$. This replaces L165–168, where the manuscript uses $\sum_{h<j} m_h \le m_j/15$. The spread statement is L196–201 with e^{-m_j} replaced by $\mathrm{e}^{-\beta m_j}$. □

Lemma 5 (Spread inflation; replaces `trees.tex` L244–252)**.** $\prod_{u<t}(1 - \mathrm{e}^{-\beta \lfloor \ell/2^u \rfloor})^{-1} \le K := \prod_{v \ge 0}(1 - \mathrm{e}^{-\beta 2^v})^{-1}$ *whenever* $\lfloor \ell/2^{t-1} \rfloor \ge 1$.

Proof. Since $\lfloor y \rfloor \ge 2^v \lfloor y/2^v \rfloor$, the v-th smallest capacity in the product is at least 2^v. The factor $(1 - \mathrm{e}^{-\beta m})^{-1}$ decreases in m. □

The manuscript bounds the product by one over *all* positive integers, giving 4 at $\beta = 1$. The dyadic product is much smaller; at $\beta = 1$ it equals 1.864.

Lemma 6 (Attempts; replaces `trees.tex` L257–277)**.** *Let $p \le 1 - B$ and $s = \lfloor \log_2 \ell_k \rfloor + 1$. Let $N(s)$ be the least N with $\mathbb{P}(\mathrm{Bin}(N, p) \ge s) \ge \frac{1}{2}$. Put $a = K\sigma$ and $\rho = \mathrm{e}^L a$. Then $W \sim \mu_r$ with $r = \min\{1, N(s)\rho\}$ contains a path union with probability at least $\frac{1}{2}$.*

Proof. By Lemma 5, every level stays a-spread throughout. So each attempt before completion succeeds with conditional probability at least p. Induction on N, conditioning on the first attempt, shows that the number of successes among N attempts stochastically dominates $\mathrm{Bin}(N, p)$. After s successes all capacities are zero. Pre-draw $N(s)$ sets; their union has law μ_{r_*} with $r_* \le N(s)\rho$. □

The manuscript uses Markov's inequality with $4s$ attempts and success probability $\frac{2}{3}$. The threshold p_c is defined with $\frac{1}{2}$, so $\frac{1}{2}$ is enough here.

3. The numbers

All quantities were evaluated in Arb ball arithmetic (256 bits) and in exact integer arithmetic (`scripts/independent/kk176_indep.py`).

- $B \le 0.395908$. We tabulate $m \le 600$ exactly. For $m > 600$ we use $\varphi_0(m)\mathrm{e}^{cS(m)} \le \mathrm{e}^{-r_1 m} + \mathrm{e}^{-r_2 m}$ with $r_1 = \alpha - \beta - c/(g-1) > 0.639$ and $r_2 = L/2 - \alpha - \beta - \log 2 - c/(g-1) > 0.221$. The coarser relaxation $S(m) \le \lfloor m/g \rfloor g/(g-1)$ gives $B \le 0.402469$, which matches `kk176_verify.py` exactly. So $p = 0.604092937329108 \le 1 - B$.

- $K \le 7.206978$ and $\lambda = 823543/46656 \le 17.651385$, so $\mathrm{e}^L K\lambda \le 59\,627.004$.
- $N(s)$ is computed exactly for $s \le 200$. Its first values are $1, 3, 5, 6, 8, 10, 11, 13$. For $s > 200$ we use $N(s) \le \lceil s/p \rceil$: every median of $\mathrm{Bin}(N, p)$ is at least $\lfloor Np \rfloor$ [2]. This gives $\sup_s N(s)/s = 5/3$, attained at $s = 3$. The worst case for $(1 + \log_2 h)$ is $h = 2^{s-1}$, so $C_1 \le 59\,627.004 \cdot \frac{5}{3} \le 99\,379$.
- For C_2 we maximise $N(s(n))/\log_2 n$, where $s(n) = \lfloor \log_2 \binom{n}{2} \rfloor + 1$. For each $s \le 400$ we evaluate it at the least n with $s(n) = s$. For $s > 400$ we use the bound $2/p + 1/200$. The supremum is at most 3.32039, so $C_2 \le 197\,985$.

If one insists on the manuscript's success probability $\frac{2}{3}$, a rigorous Chernoff/Hoeffding tail for large s gives $C_1 \le 119\,255$ and $C_2 \le 212\,769$.

4. Where the factor 10^{20} comes from

factor	manuscript	here	ratio
margin ρ/a in Lemma 3	e^{50}	$\mathrm{e}^{6.15}$	$1.10 \cdot 10^{19}$
spread multiplier σ/p_{E}	128	17.651	7.25
spread inflation	4	7.207	0.555
attempts per success	4	5/3	2.40
C_1	$2048\mathrm{e}^{50} \approx 1.062 \cdot 10^{25}$	$\le 99\,379$	$1.07 \cdot 10^{20}$

The spread inflation is worse here because $\beta = 0.4 < 1$. The smaller β pays for a smaller L. At the manuscript's own parameters our accounting gives $B = 1.234 \cdot 10^{-4}$, which is consistent with its claim $< \frac{1}{4}$.

Remark 7. Every change is either a different numerical choice or an elementary sharpening: the dyadic product, Robbins' bound, and an exact binomial count. No new idea is involved, and we did not update the Lean statement, which hard-codes $2048\mathrm{e}^{50}$ and $6144\mathrm{e}^{50}$.

References

[1] OpenAI, *The second Kahn–Kalai conjecture*, September 24, 2026, `https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-second-Kahn-Kalai-conjecture-September-24-2026`.

[2] R. Kaas and J. M. Buhrman, Mean, median and mode in binomial distributions, *Statist. Neerlandica* 34 (1980), 13–18.

[3] H. Robbins, A remark on Stirling's formula, *Amer. Math. Monthly* 62 (1955), 26–29.

[4] T. Tran, Fractional expectation thresholds and the "second" Kahn–Kalai conjecture, arXiv:2609.20546.

[5] B. Dubroff, J. Kahn, J. Park, On the "second" Kahn–Kalai conjecture, arXiv:2508.14269; and *. . . : cliques, cycles, and trees*, arXiv:2510.02005.